



Lecture 4

Data Types

Announcements

- HW 1 is due today @ 5pm
- Many students benefit from practice; sign up for **small-group tutoring**, which starts next week
 - <https://tutoring.data8.org/>

Weekly Goals

- Monday:
 - Python basics
 - Tables
 - **Today:**
 - Types of data
 - Arrays
 - Friday:
 - Creating new tables
 - Manipulating columns of tables
-

Tables Continued

(Demo)

Numbers

(Demo)

Ints and Floats

Python has two real number types

- **int**: an integer of any size
- **float**: a number with an optional fractional part

An **int** never has a decimal point; a **float** always does

A **float** might be printed using scientific notation

Three limitations of float values:

- They have limited size (but the limit is huge)
 - They have limited precision of 15-16 decimal places
 - After arithmetic, the final few decimal places can be wrong
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Strings

(Demo)

Text and Strings

A string value is a snippet of text of any length

- `'a'`
- `'word'`
- `"there can be 2 sentences. Here's the second!"`

Strings consisting of numbers can be converted to numbers

- `int('12')`
- `float('1.2')`

Any value can be converted to a string

- `str(5)`
-

Conversions

Strings that contain numbers can be converted to numbers

- `int('12')`
- `float('1.2')`
- ~~`float('one point two')`~~ # Not a good idea!

Any value can be converted to a string

- `str(5)`

Numbers can be converted to other numeric types

- `float(1)`
 - `int(1.2)` # DANGER: loses information!
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Discussion Question

Assume you have run the following statements:

```
x = 3
```

```
y = '4'
```

```
z = '5.6'
```

What's the source of the error in each example?

A. `x + y`

B. `x + int(y + z)`

C. `str(x) + int(y)`

D. `y + float(z)`

Types

(Demo)

Every value has a type

We've seen 5 types so far:

- `int: 2`
- `float: 2.2`
- `str: 'I am the eldest boy'`
- `builtin_function_or_method: abs`
- `Table`

The `type` function can tell you the type of a value

- `type(2)`
- `type(2 + 2)`

An expression's “type” is based on its value, not how it looks

- `x = 2`
 - `type(x)`
-

Arrays

(Demo)

Arrays

An array contains a sequence of values

- All elements of an array should have the same type
- Arithmetic is applied to each element individually
- Adding two arrays adds the corresponding elements
(but the arrays must be the same length!)
- A column of a table is an array

Friday: putting together arrays to make tables!
